

Use of Lagrange-multipliers

Lecture note

TET4185 - Power Markets, Resources and Environment

In this note, we will go through the application of Lagrange-multipliers connected to the solution of an optimization task (subject to constraints) and put special emphasis on the interpretation of the multipliers.

We start with an objective function - J that shall be maximized. This objective function is a function f of a set of variables $x_1, x_2, \dots, x_n$ so that:

$$\max J = \max f(x_1, x_2, \dots, x_n) \quad (1)$$

We assume maximization here, but it could be a minimization instead. Moreover, we have a constraint expressed as:

$$g(x_1, x_2, \dots, x_n) = 0 \quad (2)$$

It means that we assume an equality constraint¹.

With normal operations research notations, we express the optimization problem like this:

$$\begin{aligned} \max J &= \max f(x_1, x_2, \dots, x_n) \\ \text{s.t.: } g(x_1, x_2, \dots, x_n) &= 0 \end{aligned}$$

Where s.t. means *subject to*.

This optimization problem can be solved by defining an extended objective function, also called a Lagrange-function like this:

$$L = f(x_1, x_2, \dots, x_n) - \lambda g(x_1, x_2, \dots, x_n) \quad (3)$$

Where λ is called the Lagrange-multiplier. We can now solve the optimization by finding a stationary point of L . We notice that as long as the constraint holds, L and f will have the same value because $g = 0$.

We then proceed to find the maximum or minimum by derivation with respect to the variables, including λ , and setting the derivatives equal to zero (in principle we find a stationary point, which can be a maximum, a minimum or a saddle point):

$$\frac{\partial L}{\partial x_1} = \frac{\partial f}{\partial x_1} - \lambda \frac{\partial g}{\partial x_1} = 0$$

¹ Inequality constraints expressed for instance as: $g \geq 0$ will not be discussed in this note.

$$\begin{aligned}
\frac{\partial L}{\partial x_2} &= \frac{\partial f}{\partial x_2} - \lambda \frac{\partial g}{\partial x_2} = 0 \\
&\vdots \\
&\vdots
\end{aligned} \tag{4}$$

$$\frac{\partial L}{\partial \lambda} = g(x_1, x_2, \dots, x_n) = 0$$

This set of equations, (4), has $n+1$ equations and $n+1$ unknowns. Notice that the last equation brings us back to the constraint, equation (2). When we solve this set of equations, we get a solution with two characteristics:

- 1) The constraints are met because they are included in the equations
- 2) The solution is the optimal solution for objective function f as well, because f and L are equal. In the optimum point either $\lambda = 0$ (non-binding constraint) or $g = 0$; this is the complementary slackness condition

The Lagrange-multiplier in many cases have an economic interpretation. In power system economics, the objective is usually tied to economy: cost minimization or surplus maximization. The constraints are in most cases tied to power balance. In order to investigate the interpretation of λ , we reformulate the equation set (4) as follows:

$$\frac{\frac{\partial f}{\partial x_1}}{\frac{\partial g}{\partial x_1}} = \frac{\frac{\partial f}{\partial x_2}}{\frac{\partial g}{\partial x_2}} = \dots = \frac{\frac{\partial f}{\partial x_n}}{\frac{\partial g}{\partial x_n}} = \lambda \tag{5}$$

That means that in the optimum point, the linearized relationship between the derivative of f and g is the same for all x_i . The numerator in equation (5) represents the rate for increase of the objective function by a marginal increase of x_i . If the objective function represents the economic benefit or surplus we want to maximize, the numerator can be interpreted as the additional cost per additional unit of x_i . The denominator represents impact of a change in the constraint. Equation (5) indicates that λ represents the marginal cost for an adjustment of the constraint.

A formal way to prove that λ represents the marginal cost changing the constraint this is as follows:

We introduce a constant c in-stead of zero as the constraint:

$$g_0(x_1, x_2, \dots, x_n) = c \tag{6}$$

We then get a new optimal value of L denoted $L_0(c)$ and corresponding values of $f, x_1, x_2, \dots, \lambda$, all depending on c .

We will now find the derivative of the objective function (which we assume represent a cost) with respect to this new constraint c and we use the chain rule:

$$\frac{df_0}{dc} = \frac{\partial f_0}{\partial x_{10}} \cdot \frac{dx_{10}}{dc} + \frac{\partial f_0}{\partial x_{20}} \cdot \frac{dx_{20}}{dc} + \dots + \frac{\partial f_0}{\partial x_{n0}} \cdot \frac{dx_{n0}}{dc} \quad (7)$$

From equation (4), we see that:

$$\frac{\partial f_0}{\partial x_i} = \lambda_0 \frac{\partial g_0}{\partial x_i} \quad (8)$$

That can be included in equation (7) and we get:

$$\frac{df_0}{dc} = \lambda_0 \frac{\partial g_0}{\partial x_{10}} \cdot \frac{dx_{10}}{dc} + \lambda_0 \frac{\partial g_0}{\partial x_{20}} \cdot \frac{dx_{20}}{dc} + \dots + \lambda_0 \frac{\partial g_0}{\partial x_{n0}} \cdot \frac{dx_{n0}}{dc} = \lambda_0 \frac{dg_0}{dc} \quad (9)$$

But in this context, c is a constant and that means: $\frac{dg_0}{dc} = 1$ (10)

And consequently: $\frac{\partial f_0}{\partial c} = \lambda_0$ (11)

So the conclusion is that the Lagrange multiplier is the rate of change of the optimal value (cost) with respect to change of the constraint.

Example

Assume we are going to build a rectangular enclosure with a given area A . We want to minimize the total length of the fence. We call the length of the two sides x_1 and x_2 and the total length of the fence l . The objective function is the length:

$$f = l = 2x_1 + 2x_2 \quad (12)$$

And that shall be minimized. The constraint is:

$$g = A - x_1 x_2 = 0 \quad (13)$$

We define a Lagrange-function which in this case is:

$$L = 2x_1 + 2x_2 + \lambda(A - x_1 x_2) \quad (14)$$

First order condition for minimum is now:

$$\begin{aligned}
\frac{\partial L}{\partial x_1} &= 2 - \lambda x_2 = 0 \\
\frac{\partial L}{\partial x_2} &= 2 - \lambda x_1 = 0 \\
\frac{\partial L}{\partial \lambda} &= A - x_1 x_2 = 0
\end{aligned} \tag{15}$$

We see that the solution is $x_1 = x_2 = \sqrt{A}$, i.e. the optimal shape of the enclosure is a square. The Lagrange-multiplier is: $\lambda = 2/\sqrt{A}$. If we insert $A = (l/4)^2$ we find that $\lambda = 8/l$. In this case λ will express how much the length of the fence must increase if the area is increased by 1 m^2 .

Assume $A = 10\,000 \text{ m}^2$ which means each side is 100 m and the total length of the fence is 400 m. With these numerical values we get: $\lambda = \frac{8}{l} = \frac{8}{400} = 0.02$ which means that 1 m^2 increase of the area requires 0.02 m increase of the total fence length. This can be compared with square root of 10 001 which is:

$$\sqrt{10001} = 100.005$$

That means the length of each side has to increase by 0.005 m which means 0.02 m increase of the total length.

So the interpretation of λ in this case is the additional length (i.e. additional cost) per unit of increase of the constraint.